

Problem Setup

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Problem Setup

We are given a linear algebraic equation in terms of the variable x :

$$x + 2x + 1 = 0$$

Our objective is to solve this equation for x and find the complete set of solutions.

Strategy and Key Principles

To solve this linear equation, we apply the foundational algebraic principles of combining like terms and isolating the variable. Specifically:

1. Group all terms containing the variable x on one side of the equation.
2. Combine the like terms by adding their coefficients.
3. Isolate the variable x using inverse operations (addition and division properties of equality).

Detailed Solution

We begin with the original equation:

$$x + 2x + 1 = 0$$

Next, we combine the like terms on the left-hand side. The terms x and $2x$ can be combined by adding their coefficients (1 and 2, respectively):

$$\begin{aligned}(1 + 2)x + 1 &= 0 \\ 3x + 1 &= 0\end{aligned}$$

To isolate the term containing x , we subtract 1 from both sides of the equation:

$$\begin{aligned}3x + 1 - 1 &= 0 - 1 \\ 3x &= -1\end{aligned}$$

Finally, to solve for x , we divide both sides of the equation by the coefficient of x , which is 3:

$$\begin{aligned}\frac{3x}{3} &= \frac{-1}{3} \\ x &= -\frac{1}{3}\end{aligned}$$

Final Answer

$$x = -\frac{1}{3}$$

Key Insights

- **Combining Like Terms:** Always simplify expressions by grouping variable terms and constant terms before attempting to isolate the variable.
- **Coefficient Recognition:** Remember that a variable written without an explicit coefficient (such as x) has an implied coefficient of 1.
- **Verification:** Substituting $x = -\frac{1}{3}$ back into the original equation yields $-\frac{1}{3} + 2\left(-\frac{1}{3}\right) + 1 = -\frac{1}{3} - \frac{2}{3} + 1 = -\frac{3}{3} + 1 = -1 + 1 = 0$, confirming the solution is correct.